

A two-dimensional spiral-dilation counterexample to Conjecture 2.14 of arXiv:1812.07375

Autonomous proof audit — lane 9

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Disposition. The conjecture is false, but the negative answer is already present in the literature: Moisala’s 2020 thesis gives a geodesic L^1 example, and Moisala–Pasqualetto explicitly point to it. The proof below records a simpler finite-dimensional counterexample obtained from the rotation template of Moisala–Pasqualetto. No novelty claim is made.

1 Source question and transcription

The source paper asks (Conjecture 2.14, printed page 12):

If G is a metric scalable group that is separable and nilpotent and its distance is geodesic, then its first layer $V_1(G)$ generates G .

Here a scalable group has automorphisms δ_t for $t \neq 0$, with $\delta_t \delta_s = \delta_{ts}$, a continuous map $(t, x) \mapsto \delta_t x$, and $\delta_0 x = e$. A compatible left-invariant metric satisfies

$$d(\delta_t x, \delta_t y) = |t| d(x, y).$$

The first layer is

$$V_1(G) = \{x : \delta_{t+s} x = \delta_t x \delta_s x \text{ for every } s, t \in \mathbb{R}\}.$$

“Generates” means that the closed subgroup generated by its dilates is all of G .

2 Idea and proof intuition

Metric homogeneity determines the radial size of a dilation but not its angular part. Let the angular part rotate logarithmically with the scale. The logarithm converts multiplication of parameters into addition of angles, so the scalable-group law remains true. The orbit of any nonzero vector then spirals as it contracts. Such an orbit cannot be a one-parameter subgroup for vector addition, and in fact the single test $s = t = 1$ already excludes every nonzero vector from the first layer.

3 Counterexample

For $a \in \mathbb{R}$, let R_a be counterclockwise rotation of $\mathbb{R}^2$ through angle a . Take $G = (\mathbb{R}^2, +)$, the Euclidean metric, and define

$$\delta_0 x = 0, \quad \delta_t x = t R_{\log|t|} x \quad (t \neq 0).$$

Proof of Proposition 1.8. Assume first that $(N_m)_m$ is a filtration by Carnot subgroups of G . Then the set $\cup_m V_1(N_m) \subseteq V_1(G)$ and it generates $\cup_m N_m$. As $\cup_m N_m$ is dense in G , we get that $V_1(G)$ generates G as a scalable group. Assume then that $V_1(G)$ generates and let $(a_n)_{n \in \mathbb{N}} \subseteq V_1(G)$ be a dense sequence in $V_1(G)$. This sequence generates a dense subgroup of G , and choosing N_m to be the scalable group generated by $\{a_1, \dots, a_m\}$ gives G a filtration by Carnot groups by Proposition 2.1. $\square$

It would be interesting to find geometric conditions that would allow us to conclude that $V_1(G)$ generates G . Indeed, in the case of simply connected nilpotent Lie groups admitting dilations, geodcity then implies that the first layer generates the entire group as rectifiable curves can be approximated by horizontal line segments. Thus, we make the following conjecture.

Conjecture 2.14. *If G is a metric scalable group that is separable and nilpotent and its distance is geodesic, then its first layer $V_1(G)$ generates G .*

3. NEGLIGIBLE SETS OF METRIC SCALABLE GROUPS

In this section (G, d, δ) denotes a metric scalable group (according to Definition 1.2).

Figure 1: Source-page evidence: arXiv:1812.07375, printed page 12.

Theorem 1. *The triple (G, δ, d_2) is a complete separable geodesic metric scalable group. Its underlying group is abelian, hence nilpotent, but $V_1(G) = \{0\}$. Consequently $V_1(G)$ does not generate G .*

Proof. For $t \neq 0$, δ_t is an invertible linear map and therefore a group automorphism. If $s, t \neq 0$, commutativity of planar rotations and $\log |st| = \log |s| + \log |t|$ give

$$\begin{aligned} \delta_t(\delta_s x) &= t R_{\log |t|} (s R_{\log |s|} x) \\ &= st R_{\log |st|} x = \delta_{st} x. \end{aligned}$$

The identities with $st = 0$ follow from $\delta_0 x = 0$. For $t \neq 0$,

$$\|\delta_t x - \delta_t y\|_2 = |t| \|x - y\|_2,$$

because rotations are isometries. This also proves continuity at $t = 0$: if $(t_n, x_n) \rightarrow (0, x)$, then $\|\delta_{t_n} x_n\|_2 = |t_n| \|x_n\|_2 \rightarrow 0$. Joint continuity away from zero is immediate from the formula. Thus all scalable-group and metric-compatibility axioms hold, including for negative parameters.

Euclidean $\mathbb{R}^2$ is complete, separable, and geodesic, and its additive group is abelian. Finally, if $x \in V_1(G)$, the defining identity with $s = t = 1$ would imply

$$2R_{\log 2} x = \delta_2 x = \delta_1 x + \delta_1 x = 2x.$$

Since $0 < \log 2 < 2\pi$, the nontrivial rotation $R_{\log 2}$ has no nonzero fixed vector. Hence $x = 0$. Conversely $0 \in V_1(G)$, so $V_1(G) = \{0\}$ and its generated closed subgroup is proper. $\square$

4 Literature status and verification

The original conjecture is already answered negatively. On page 21 of the introductory part of her thesis, Moisala constructs a geodesic example on the additive group $L^1(\mathbb{R})$ with $(\delta_\lambda f)(x) = f(x/\lambda)$

and proves that its first layer is trivial. Moisala–Pasqualetto later state explicitly that an abelian complete metric scalable group with trivial first layer “can even be made geodesic”, citing that thesis page.

Their Remark 3.4 also uses the general rotation template $\delta_t = (tI)R_{f(t)}$ for a multiplicative-to-additive homomorphism f . Choosing the continuous homomorphism $f(t) = \log |t|$ gives precisely the counterexample above. The argument has been checked against every axiom in the source definition; the test $s = t = 1$ independently certifies the trivial first layer. Local compactness causes no contradiction: metric characterizations may identify the underlying metric group as a Carnot group without forcing an arbitrarily specified scalable structure to be its standard Carnot dilation family.

Limitations and novelty. This packet refutes only the printed conjecture. It does not classify which extra compatibility condition on a given dilation family forces generation by the first layer. The negative answer is prior literature. The two-dimensional presentation is an independently derived simplification of a published rotation template, not asserted to be new.

References

- [1] E. Le Donne, S. Li, and T. Moisala, *Gâteaux differentiability on infinite-dimensional Carnot groups*, arXiv:1812.07375, Conjecture 2.14 (2018).
- [2] T. Moisala, *Unraveling Intrinsic Geometry of Sets and Functions in Carnot groups*, JYU Dissertations 264, University of Jyväskylä (2020), introductory p. 21, <https://jyx.jyu.fi/handle/123456789/71351>.
- [3] T. Moisala and E. Pasqualetto, *Direct limits of infinite-dimensional Carnot groups*, arXiv:2101.03979, Remarks following Proposition 4.5 and Remark 3.4 (2021).